Checkpoint 6 Maths — Practice Test 1 — Mark Scheme

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 40 marks

1 The school library counts its books. It has 2 thousands, 15 hundreds, 3 tens and 8 ones. **[1]**

How many books does the library have?

Answer: 3538
Notes / working: Regroup: 15 hundreds = 1 thousand and 5 hundreds.
2000 + 1500 + 30 + 8 = **3538** books.
Common mistake: writing 21 538 by putting the digits 2, 15, 3, 8 side by side.

2 A sequence counts back in steps of the same size. **[1]**

2.0, 1.4, 0.8, _____, _____

Write the next two numbers. Type them with a comma between them, for example 5, 3

Answer: 0.2, -0.4 or 0.2,-0.4 or 0.2, −0.4 or 0.2 and -0.4 or 0.2 and −0.4 or .2, -.4
Notes / working: The step is 2.0 − 1.4 = 0.6, so subtract 0.6 each time.
0.8 − 0.6 = **0.2**
0.2 − 0.6 = **−0.4** (go past zero: 0.2 takes you to 0, then 0.4 more below zero).
Both numbers are needed for the mark.

3 **[1]**

(a) Work out 30 − 4 × 6 ÷ 3

Answer: 22
Notes / working: Multiplication and division come before subtraction, working from left to right.
4 × 6 = 24, then 24 ÷ 3 = 8.
30 − 8 = **22**.
Common mistake: working left to right from the start: 30 − 4 = 26, × 6 = 156, ÷ 3 = 52.

(b) Write one pair of brackets to make this calculation correct. **[1]**

5 + 3 × 4 − 2 = 30

> **Answer:** 1 mark: (5 + 3) × 4 − 2 = 30. Accept ((5 + 3) × 4) − 2. Do not accept any other position of the brackets.
> **Notes / working:** Try the brackets in different places:
> (5 + 3) × 4 − 2 = 8 × 4 − 2 = 32 − 2 = **30** ✓
> 5 + 3 × (4 − 2) = 5 + 6 = 11 ×
> (5 + 3) × (4 − 2) uses two pairs, and gives 16 ×
> Without brackets: 5 + 12 − 2 = 15 ×
> So the brackets go around **5 + 3**.

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4 A juice stall sells 1245 cups of juice in April. **[2]**

Each cup costs 24 pounds.

How much money does the stall take for these cups in April? Show your working.

> **Answer:** 29880 pounds
> **Notes / working:** Estimate first: 1200 × 25 = 30 000, so the answer is about 30 000.
> 1245 × 20 = 24 900
> 1245 × 4 = 4980
> 24 900 + 4980 = **29 880 pounds**
> Mark scheme: M1 for a complete correct method (for example the two partial products 24 900 and 4980, or a grid method) with no more than one arithmetic error; A1 for 29 880.

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5 The juice stall has 912 oranges. The oranges are packed into bags of 16. **[2]**

How many bags can be filled? Show your working.

> **Answer:** 57 bags
> **Notes / working:** Estimate: 900 ÷ 15 = 60, so the answer is a little under 60.
> 16 × 50 = 800, leaving 912 − 800 = 112.
> 16 × 7 = 112, leaving 0.
> 50 + 7 = **57 bags** (exactly, no oranges left over).
> Check: 57 × 16 = 570 + 342 = 912 ✓
> Mark scheme: M1 for a correct method (chunking, short or long division) with at most one error; A1 for 57.

6 On a class trip to the zoo, the sea-lion show starts every 15 minutes. **[1]**
The parrot show starts every 20 minutes.
Both shows start at 10:00

What is the next time that both shows start together?

Answer: 11:00 or 11.00 or 11 am or 11:00 am or 11 o'clock or 1100
Notes / working: List the start times (minutes after 10:00).
Sea lions: 15, 30, 45, **60**, …
Parrots: 20, 40, **60**, …
The first common multiple of 15 and 20 is 60. 60 minutes after 10:00 is **11:00**.

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7 **[1]**

Which of these numbers is divisible by 9?

- ☐ A) 1236
- ☐ B) 4517
- ☐ C) 6024
- ☐ D) 2754

Answer: D) 2754
Notes / working: A number is divisible by 9 when the sum of its digits is divisible by 9.
1236 → 1 + 2 + 3 + 6 = 12 × (divisible by 3 only)
4517 → 17 ×
6024 → 12 ×
2754 → 2 + 7 + 5 + 4 = 18 ✓
So **2754** (D). Check: 2754 ÷ 9 = 306.

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8 **[1]**

Which of these is a cube number?

- ☐ A) 16
- ☐ B) 27
- ☐ C) 49
- ☐ D) 50

Answer: B) 27
Notes / working: Cube numbers: 1 × 1 × 1 = 1, 2 × 2 × 2 = 8, 3 × 3 × 3 = **27**, 4 × 4 × 4 = 64, 5 × 5 × 5 = 125.
16 and 49 are square numbers, not cube numbers. 50 is neither. Answer B.

9 In this sequence, each term is the position number squared, then subtract 1. **[1]**

Position	1	2	3	4
Term	0	3	8	15

What is the 9th term?

Answer: 80
Notes / working: 9th term = $9^2 - 1 = 81 - 1 =$ **80**.
Check the rule on the table: $4^2 - 1 = 16 - 1 = 15$ ✓

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10 Pupils make a pattern of books on the library display shelves. **[1]**

Position	1	2	3	4
Number of books	7	14	21	28

(a) How many books are there in position 20?

Answer: 140 books
Notes / working: Adding 7 each time is the same as multiplying the position by 7.
Rule: number of books = position × 7.
20 × 7 = **140**.
Common mistake: counting on 7 only once more (28 + 7 = 35), or doubling the 10th term from a wrong rule.

(b) Sara says, "One of the positions has exactly 150 books." **[1]**
Is Sara correct? Explain how you know.

Answer: 1 mark: No, with a correct reason, e.g. 150 is not a multiple of 7 / 150 ÷ 7 is not a whole number / 7 × 21 = 147 and 7 × 22 = 154, so 150 is between two terms. Do not accept "No" with no reason or an incorrect reason.
Notes / working: Every term is position × 7, so every term is a multiple of 7.
7 × 21 = 147 and 7 × 22 = 154. 150 is between them, so it is **not** a term. Sara is **not correct**.

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11 On the zoo trip there are b boys and g girls. **[1]**
b + g = 42

b = 19. What is the value of g?

Answer: 23
Notes / working: b + g = 42 and b = 19, so 19 + g = 42.
g = 42 − 19 = **23**.
The letters stand for numbers that can change; here the total stays 42.

12 At the zoo, the temperature inside the penguin house is −3 °C. **[1]**
The temperature inside the reptile house is 28 °C.

What is the difference between the two temperatures?

> **Answer:** 31 °C
> **Notes / working:** From −3 up to 0 is 3 degrees. From 0 up to 28 is 28 degrees.
> 3 + 28 = **31 °C**.
> Common mistake: 28 − 3 = 25 (forgetting that −3 is below zero).

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13 **[1]**

What is the value of the digit 6 in the number 4.316?

> **Answer:** 0.006 or 6/1000
> **Notes / working:** 4.316 = 4 ones + 3 tenths + 1 hundredth + 6 thousandths.
> The 6 is in the thousandths column, so its value is **0.006** (6 thousandths).

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14 **[1]**

Work out 45 ÷ 1000

- ☐ A) 0.45
- ☐ B) 4.5
- ☐ C) 0.045
- ☐ D) 0.0045

> **Answer:** C) 0.045
> **Notes / working:** Dividing by 1000 moves every digit three places to the right (smaller).
> 45 → 4.5 (÷ 10) → 0.45 (÷ 100) → **0.045** (÷ 1000). Answer C.

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15 **[1]**

(a) Ola says, "20/32 written in its simplest form is 10/16."
Is Ola correct? Tick (✓) Yes or No and explain your answer.

> **Answer:** 1 mark: No, with a correct explanation, e.g. 10 and 16 still have a common factor of 2 / 10/16 can be simplified again / the simplest form is 5/8. Do not accept "No" with no explanation.
> **Notes / working: No.** 20/32 = 10/16 is correct, but 10/16 is **not** in its simplest form: 10 and 16 still share the factor 2.
> 10/16 = 5/8, and 5 and 8 have no common factor except 1, so the simplest form is **5/8**.
> Quick way: divide by the highest common factor of 20 and 32, which is 4.

(b) Work out 3/5 of 45 **[1]**

> **Answer:** 27
> **Notes / working:** 1/5 of 45 = 45 ÷ 5 = 9.
> 3/5 of 45 = 9 × 3 = **27**.

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16 In the school library, 2/5 of the books are fiction and 1/3 of the books are non-fiction. The rest of the books are magazines. **[2]**

What fraction of the books are magazines? Show your working and give your answer in its simplest form.

> **Answer:** 4/15
> **Notes / working:** Use a common denominator of 15.
> 2/5 = 6/15 and 1/3 = 5/15.
> Fiction and non-fiction: 6/15 + 5/15 = 11/15.
> Magazines: 15/15 − 11/15 = **4/15**.
> Mark scheme: M1 for 2/5 and 1/3 written with a common denominator (6/15 and 5/15) or 11/15 seen; A1 for 4/15 (oe).

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17 **[1]**

Work out 2/9 × 4

> **Answer:** 8/9
> **Notes / working:** 2/9 × 4 means 4 lots of 2/9: multiply the numerator by 4.
> 2 × 4 = 8, so the answer is **8/9**. (The denominator stays 9.)

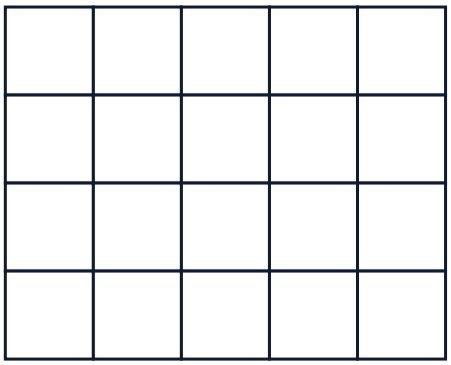
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18 **[1]**

Write <, > or = in each box.
0.7 ☐ 3/4
40% ☐ 2/5

> **Answer:** 1 mark for both correct: 0.7 < 3/4 and 40% = 2/5.
> **Notes / working:** Change everything to decimals (or percentages).
> 3/4 = 0.75, and 0.7 = 0.70 is less than 0.75, so **0.7 < 3/4**.
> 2/5 = 4/10 = 0.4 = 40%, so **40% = 2/5**.

19 **[1]**



Shade 35% of this grid.

Answer: 1 mark: exactly 7 of the 20 squares shaded (any 7 squares).
Notes / working: The grid has 5 × 4 = 20 squares, so each square is 100% ÷ 20 = 5%.
35% ÷ 5% = 7, so shade **7 squares** (any 7).
Check: 7/20 = 35/100 = 35%.

20 **[1]**

Work out 7.3 − 2.86

Answer: 4.44
Notes / working: Line up the decimal points and write 7.3 as 7.30.
7.30 − 2.86 = **4.44**.
Check: 4.44 + 2.86 = 7.30 ✓
Common mistake: 7.3 − 2.86 = 5.57 (lining the digits up from the right).

21 The juice stall makes mango-orange juice. **[1]**
It mixes mango juice and orange juice in the ratio 2 : 5

The stall uses 6 cups of mango juice. How many cups of orange juice does it use?

Answer: 15 cups
Notes / working: 2 : 5 is equivalent to 6 : ?
2 × 3 = 6, so multiply both parts by 3: 5 × 3 = **15 cups** of orange juice.
Ratio 2 : 5 = 6 : 15.

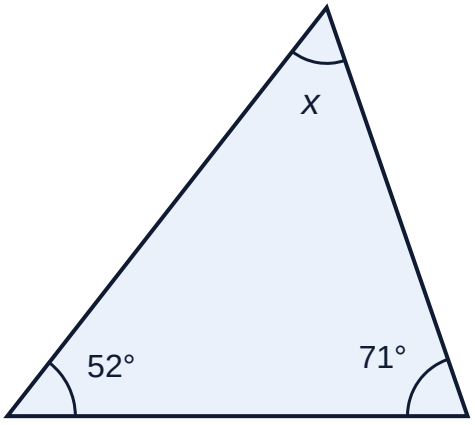
22 The class trip to the zoo lasts 3.75 hours. **[1]**

3.75 hours = 3 hours and how many minutes?

> **Answer:** 45 minutes
> **Notes / working:** 0.75 of an hour = 3/4 of 60 minutes = **45 minutes**.
> So 3.75 hours = 3 hours 45 minutes.
> Common mistake: 3 hours 75 minutes — the digits after the point are a decimal part of an hour, not minutes.

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23 **[1]**



Work out the size of angle x.

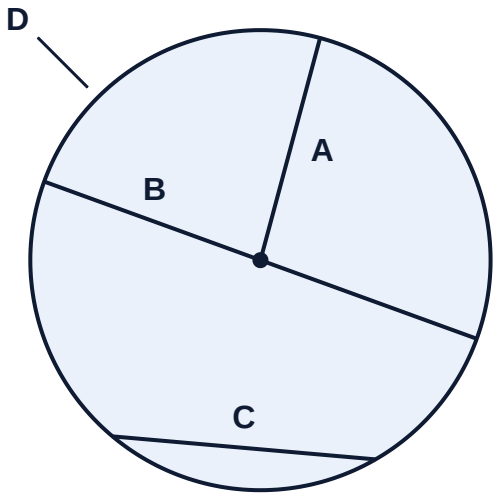
> **Answer:** 57 °
> **Notes / working:** The angles in a triangle add up to 180°.
> 52° + 71° = 123°
> x = 180° − 123° = **57°**.

24 **[1]**

Sketch a quadrilateral that has exactly one pair of parallel sides and two right angles.
Write the name of your shape.

Answer: 1 mark: a sketch of a right-angled trapezium (one pair of parallel sides, two right angles, the other two sides not parallel) AND the name trapezium. Accept recognisable misspellings. Do not accept a square, rectangle or parallelogram.
Notes / working: A **trapezium** has exactly one pair of parallel sides. If the side joining the two parallel sides is at right angles to them, the trapezium has two right angles (a right-angled trapezium).
Sketch: draw a long horizontal bottom side, a shorter horizontal top side starting above its left end, a vertical left side, and a slanting right side.
A rectangle has two pairs of parallel sides and four right angles, so it is not correct.

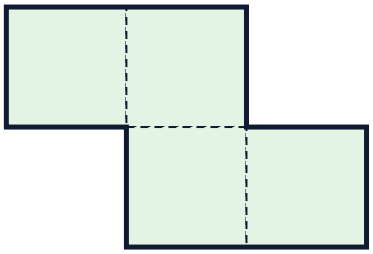
25 Look at the circle. The dot is the centre. **[1]**



Which letter shows a radius of the circle?

Answer: A
Notes / working: A radius goes from the centre to the circumference: that is line **A**.
B passes through the centre from one side to the other, so it is a diameter. C does not pass through the centre (it is a chord). D points to the circumference (the curved edge).

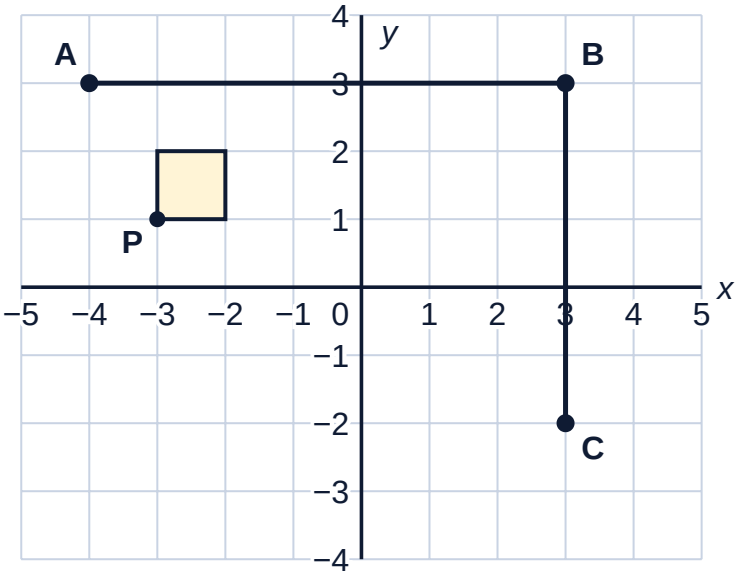
26 A garden path is made from paving slabs in this shape. **[1]**



What is the order of rotational symmetry of the shape?

Answer: 2
Notes / working: Trace the shape and turn the tracing around its centre. It fits on top of itself after a half turn (180°) and again after a full turn.
So it fits **2** times in a full turn: rotational symmetry of **order 2**.
Note: the shape has no lines of symmetry, but it still has rotational symmetry.

27 A garden plot plan is drawn on a coordinate grid. A, B and C are three corners of a rectangular plot. The square is a shed. P is one corner of the shed. **[1]**



(a) Write the coordinates of point A.

Answer: (-4, 3) or (-4,3) or -4,3 or (−4, 3)
Notes / working: Go across first (x), then up or down (y). A is 4 to the left of 0 and 3 up, so A = **(−4, 3)**.

(b) D is the fourth corner of the rectangle ABCD. Write the coordinates of D. **[1]**

> **Answer:** (-4, -2) or (-4,-2) or -4,-2 or (−4, −2)
> **Notes / working:** In a rectangle, D is below A and level with C.
> D has the same x-coordinate as A (−4) and the same y-coordinate as C (−2).
> D = **(−4, −2)**.

(c) The shed is moved 5 squares to the right and 2 squares down. Write the new coordinates of corner P. **[1]**

> **Answer:** (2, -1) or (2,-1) or 2,-1 or (2, −1)
> **Notes / working:** P = (−3, 1).
> 5 to the right: x = −3 + 5 = 2.
> 2 down: y = 1 − 2 = −1.
> New P = **(2, −1)**.

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28 A bag has 6 red counters and 2 blue counters. Ali takes one counter without looking. **[1]**

Which word best describes the chance that Ali takes a blue counter?

- ☐ A) unlikely
- ☐ B) impossible
- ☐ C) even chance
- ☐ D) likely

> **Answer:** A) unlikely
> **Notes / working:** 2 of the 8 counters are blue: 2/8 = 1/4. That is less than an even chance (1/2), but it can happen.
> So it is **unlikely** (A). It is not impossible, because there are blue counters in the bag.

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29 A spinner has eight equal sections numbered 1 to 8. Rana spins it once. **[1]**

Explain why the events "the number is odd" and "the number is a multiple of 4" are mutually exclusive.

> **Answer:** 1 mark: an explanation that the two events cannot happen at the same time because no number on the spinner is both odd and a multiple of 4, e.g. the multiples of 4 are 4 and 8, which are both even. Do not accept just "they cannot happen together" with no reference to the numbers.
> **Notes / working:** Odd numbers: 1, 3, 5, 7. Multiples of 4: 4, 8.
> No number is in both lists (all multiples of 4 are even), so the two events **cannot happen at the same time**. Events like this are called mutually exclusive.

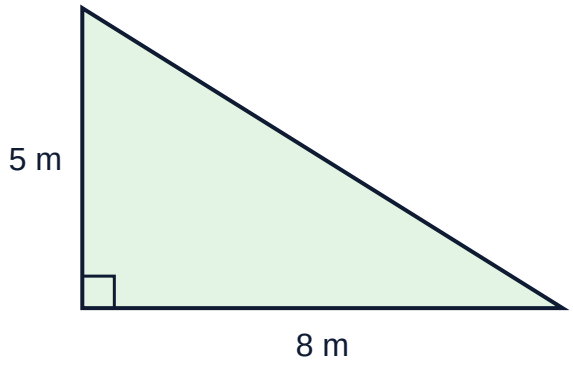
30 Seven pupils write down how many library books they borrowed this term. **[1]**

3, 5, 2, 5, 8, 4, 1

What is the median number of books?

Answer: 4
Notes / working: Put the numbers in order: 1, 2, 3, **4**, 5, 5, 8.
There are 7 numbers, so the median is the 4th one: **4**.
Common mistake: giving 5 (that is the mode) or taking the middle of the unordered list (5).

31 A flower bed in the garden is a right-angled triangle. **[1]**



Work out the area of the flower bed.

Answer: 20 m^2
Notes / working: A right-angled triangle is half of a rectangle.
Rectangle: 8 × 5 = 40 m².
Triangle: 40 ÷ 2 = **20 m²**.

32 Omar says, "Every number that is divisible by 3 is also divisible by 6." **[1]**

Give an example to show that Omar is wrong. Explain your example.

Answer: 1 mark: an odd multiple of 3 (e.g. 3, 9, 15, 21, 27, 33…) with a reason, e.g. 15 ÷ 3 = 5 but 15 is odd so it is not divisible by 6 / 15 ÷ 6 = 2.5. Do not accept an example that is a multiple of 6.
Notes / working: To be divisible by 6, a number must be divisible by 3 **and** be even.
Example: **9**. 9 ÷ 3 = 3, but 9 is odd, so 9 ÷ 6 = 1.5 is not a whole number.
One counter-example is enough to show that a statement is wrong.

Paper 2 — 45 minutes — 40 marks

1 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[1]**

Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

(a) Round Seif's time to the nearest tenth of a second.

Answer: 15.7 seconds
Notes / working: 15.66: the tenths digit is 6; look at the hundredths digit, 6. It is 5 or more, so round up.
15.66 → **15.7 seconds**.

(b) What is the mode of the times? **[1]**

Answer: 14.87 seconds
Notes / working: The mode is the value that appears most often. 14.87 appears twice (Karim and Ziad).
Mode = **14.87 seconds**.

(c) What is the median time? **[1]**

Answer: 15.42 seconds
Notes / working: Order the times: 14.87, 14.87, 15.20, **15.42**, 15.66, 15.80, 16.05.
The middle (4th) value is **15.42 seconds**.

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2 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[2]**

Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

Work out the mean time. Show your working.

Answer: 15.41 seconds
Notes / working: Total = 15.42 + 14.87 + 16.05 + 15.20 + 14.87 + 15.80 + 15.66 = 107.87
Mean = 107.87 ÷ 7 = **15.41 seconds**.
Mark scheme: M1 for 107.87 ÷ 7 (or the total 107.87 seen); A1 for 15.41.

3 At sports day, seven pupils from Class 6A run 100 m. The table shows their times. **[2]**

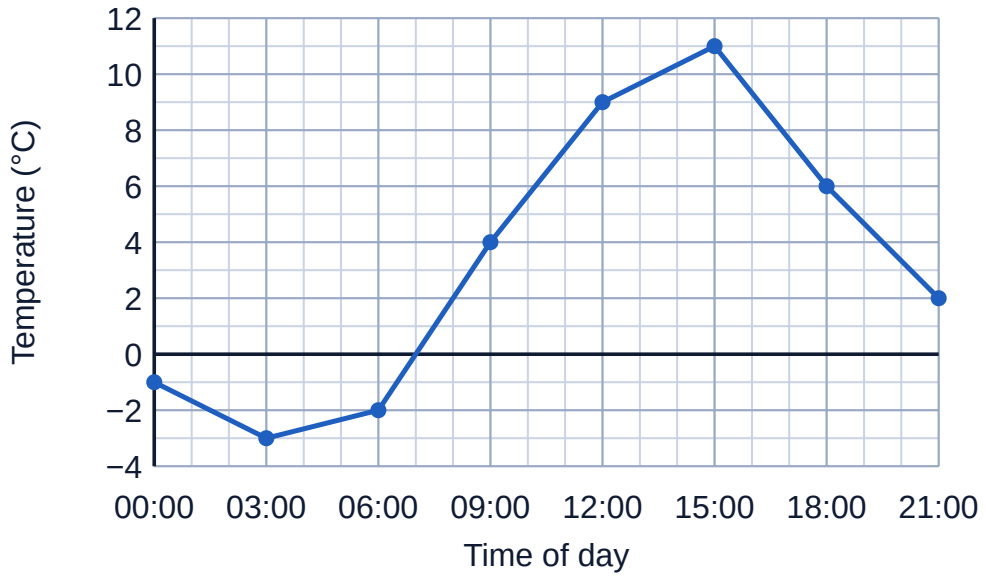
Name	Adam	Karim	Youssef	Omar	Ziad	Hamza	Seif
Time (seconds)	15.42	14.87	16.05	15.20	14.87	15.80	15.66

For Class 6B the mean time is 15.90 seconds and the range is 0.62 seconds.

Compare the times of Class 6A and Class 6B. Write two different comparisons.

Answer: 1 mark: Class 6A were faster on average, because their mean is lower (15.41 < 15.90) — FT their mean from Question 2. 1 mark: Class 6B's times were more consistent / closer together / less spread out, because their range is smaller (0.62 < 1.18); accept the range of 6A worked out as 1.18. Do not accept "6B is faster because 15.90 is bigger". Each comparison must refer to the mean or the range.
Notes / working: Mean: 6A = 15.41 s, 6B = 15.90 s. A **lower** time is faster, so **6A were faster on average**.
Range: 6A = 16.05 − 14.87 = 1.18 s; 6B = 0.62 s. 6B's range is smaller, so **6B's times were more consistent** (closer together).
Exam tip: compare one average AND one measure of spread, and say what each one means in the context.

4 The weather chart shows the temperature in a town in Sinai, measured every 3 hours on one day in January. **[1]**



(a) What was the temperature at 09:00?

Answer: 4 °C
Notes / working: Go up from 09:00 to the line, then across to the temperature axis: **4 °C**.

(b) What is the difference between the highest and the lowest temperature on the chart? **[1]**

> **Answer:** 14 °C
> **Notes / working:** Highest: 11 °C (at 15:00). Lowest: −3 °C (at 03:00).
> From −3 up to 0 is 3, from 0 up to 11 is 11: 3 + 11 = **14 °C**.
> Common mistake: 11 − 3 = 8.

(c) Use the graph to estimate the temperature at 08:00. **[1]**

> **Answer:** 2 °C
> **Notes / working:** Each small square across is 1 hour. 08:00 is two small squares after 06:00. Between 06:00 (−2 °C) and 09:00 (4 °C) the temperature rises 6 degrees in 3 hours, so 2 degrees each hour.
> At 08:00: −2 + 2 + 2 = **2 °C** (accept 1.5 to 2.5).

(d) Explain why a line graph is a good way to show this data. **[1]**

> **Answer:** 1 mark: temperature is continuous data / it changes all the time (between the readings), so joining the points shows how it changes over time and lets you read values between the times. Do not accept "it is easy to read" alone.
> **Notes / working:** Temperature is **continuous** data: it changes all the time, not only at 00:00, 03:00, … A line graph joins the readings, so it shows the **change over time** and lets you estimate temperatures between the readings (as in part (c)).

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5 The juice stall shows this price list. **[2]**

Juice	Price of one cup
Orange	18.50 pounds
Mango	22.75 pounds
Guava	15.25 pounds

Huda buys 3 cups of mango juice and 2 cups of guava juice. She pays with a 200-pound note. How much change does she get? Show your working.

> **Answer:** 101.25 pounds
> **Notes / working:** Mango: 3 × 22.75 = 68.25
> Guava: 2 × 15.25 = 30.50
> Total: 68.25 + 30.50 = 98.75
> Change: 200 − 98.75 = **101.25 pounds**.
> Mark scheme: M1 for 98.75 seen (or 200 − 3 × 22.75 − 2 × 15.25); A1 for 101.25. Accept 101.25 written as 101 pounds 25 piastres.

6 **[1]**

A jug holds 2.4 litres of orange juice. It is poured equally into 8 cups.
How much juice is in each cup?

Answer: 0.3 litres
Notes / working: 2.4 ÷ 8 = **0.3 litres** (that is 300 ml).
Check: 0.3 × 8 = 2.4 ✓

7 **[1]**

The stall uses 6 mangoes to make 4 cups of mango juice.
How many mangoes does it need to make 18 cups?

Answer: 27 mangoes
Notes / working: The number of mangoes and the number of cups are in direct proportion.
4 cups need 6 mangoes, so 2 cups need 3 mangoes (halve both).
18 cups = 9 × 2 cups, so 9 × 3 = **27 mangoes**.
(Or: 6 ÷ 4 = 1.5 mangoes per cup; 18 × 1.5 = 27.)

8 **[1]**

On Friday the stall sells 240 cups of juice. 35% of them are orange juice.
How many cups of orange juice does it sell?

Answer: 84 cups
Notes / working: 10% of 240 = 24, so 30% = 72. 5% = 12.
35% = 72 + 12 = **84 cups**.
(Calculator: 240 × 35 ÷ 100 = 84.)

9 152 pupils and 9 teachers go on a class trip to the zoo. **[1]**

(a) Each bus can carry 48 people. What is the smallest number of buses they need?

Answer: 4 buses
Notes / working: Number of people = 152 + 9 = 161.
161 ÷ 48 = 3.35…
3 buses carry only 144 people, so they need **4 buses**. Always round up when everyone must have a seat.
Common mistake: answering 3 or 3.35.

10 152 pupils and 9 teachers go on a class trip to the zoo. **[2]**
Zoo tickets cost 85 pounds for each pupil and 120 pounds for each teacher.

(b) Work out the total cost of the tickets. Show your working.

Answer: 14000 pounds
Notes / working: Pupils: 152 × 85 = 12 920
Teachers: 9 × 120 = 1080
Total: 12 920 + 1080 = **14 000 pounds**.
Mark scheme: M1 for 152 × 85 + 9 × 120 (or 12 920 or 1080 seen with an attempt to add); A1 for 14 000.

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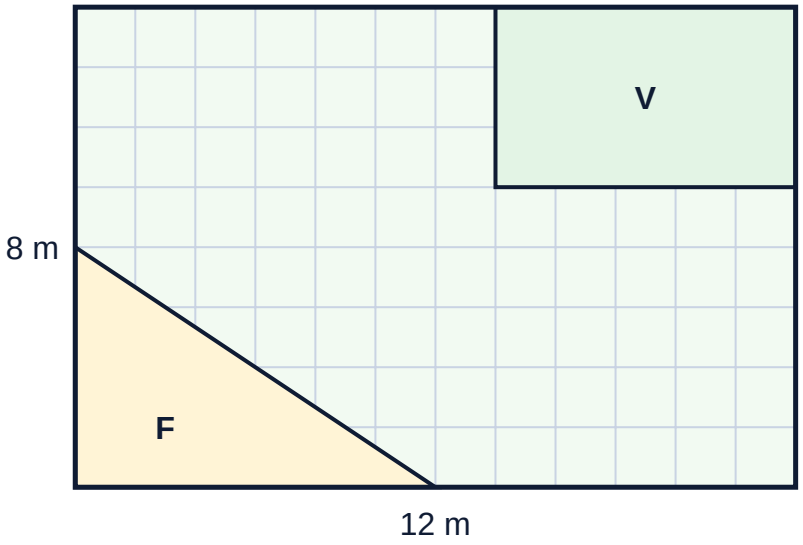
11 **[1]**

The bus journey home takes 1.4 hours.
1.4 hours = 1 hour and how many minutes?

Answer: 24 minutes
Notes / working: 0.4 of an hour = 0.4 × 60 = **24 minutes**.
So 1.4 hours = 1 hour 24 minutes (not 1 hour 40 minutes).

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12 The plan shows a rectangular school garden plot on a square grid. Each square is 1 m by 1 m. **[1]**



F = flower bed, V = vegetable bed, the rest is grass

(a) Work out the area of the flower bed F.

Answer: 12 m^2
Notes / working: Count squares: the flower bed is a right-angled triangle 6 m along the bottom and 4 m up the side.
Area = (6 × 4) ÷ 2 = 24 ÷ 2 = **12 m²**.

(b) Work out the area of the grass. Show your working. **[2]**

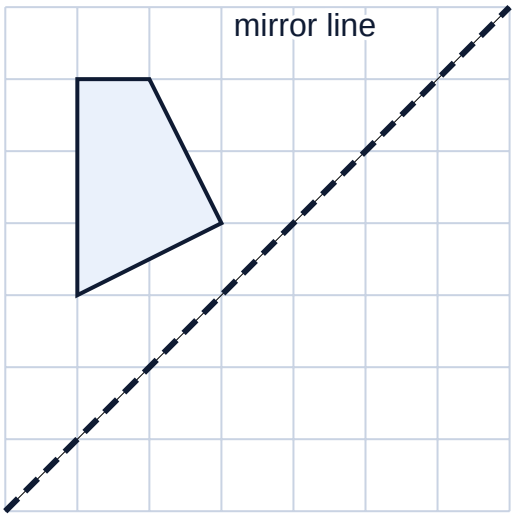
Answer: 69 m^2
Notes / working: Whole garden: 12 × 8 = 96 m².
Flower bed: 12 m² (from part a).
Vegetable bed: 5 × 3 = 15 m².
Grass: 96 − 12 − 15 = **69 m²**.
Mark scheme: M1 for 96 − (their 12) − 15, or 96 and 15 both seen; A1 for 69.

(c) The vegetable bed is shared between tomatoes and beans in the ratio 3 : 2 **[1]**
What area is used for tomatoes?

Answer: 9 m^2
Notes / working: Vegetable bed = 15 m². The ratio 3 : 2 has 3 + 2 = 5 parts.
One part = 15 ÷ 5 = 3 m².
Tomatoes = 3 parts = 3 × 3 = **9 m²** (beans = 6 m²).

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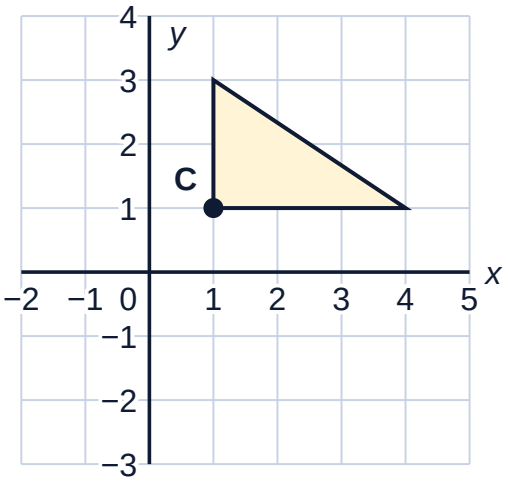
13 **[1]**



Draw the reflection of the shape in the mirror line.

Answer: 1 mark: the correct reflection drawn. Counting squares from the bottom-left corner of the grid (across, up), the image has its vertices at (3, 1), (6, 1), (6, 2) and (4, 3). Accept a correct reflection drawn freehand if the vertices are clearly at these grid points.
Notes / working: The mirror line is diagonal. For each corner, count squares to the mirror line at right angles to it (diagonally) and go the same distance on the other side.
A quick way on this grid: label the bottom-left corner (0, 0). The shape's corners are (1, 3), (1, 6), (2, 6), (3, 4). Reflecting in this diagonal swaps the two numbers: (3, 1), (6, 1), (6, 2), (4, 3).
Join these points. The image should be the same size and shape, and flipped over the line.

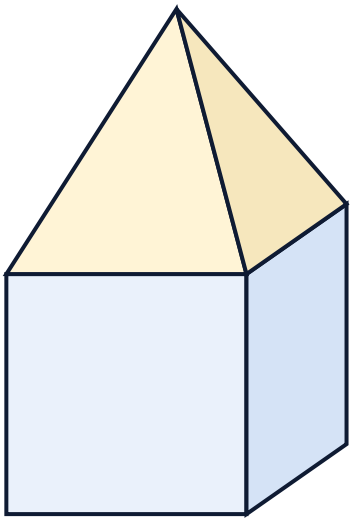
14 **[1]**



Rotate the triangle 90° clockwise about the vertex C.
Draw the new triangle on the grid.

Answer: 1 mark: triangle drawn with vertices at (1, 1), (1, −2) and (3, 1). Do not accept a 90° anticlockwise rotation (vertices (1, 1), (1, 4), (−1, 1)).
Notes / working: C = (1, 1) stays where it is.
The side from C to (4, 1) goes 3 squares right. After a quarter turn clockwise it goes 3 squares **down**: to (1, −2).
The side from C to (1, 3) goes 2 squares up. After a quarter turn clockwise it goes 2 squares **right**: to (3, 1).
New triangle: **(1, 1), (1, −2), (3, 1)**. Tip: use tracing paper, hold the pencil point on C and turn the paper a quarter turn clockwise.

15 This shape is made from a cube with a square-based pyramid on top. **[1]**



How many faces does the shape have?

Answer: 9 faces
Notes / working: The cube has 6 faces, but its top face is covered by the pyramid: 5 faces show.
The pyramid has 5 faces, but its square base is on the cube: 4 triangles show.
5 + 4 = **9 faces**.
Common mistake: 6 + 5 = 11 (counting the two hidden faces where the shapes join).

16 **[1]**

Sketch a net of a square-based pyramid.

Answer: 1 mark: one square and four triangles, arranged so that they fold into a square-based pyramid (e.g. a triangle on each side of the square). Do not accept a net with the wrong number of faces or with two triangles on the same side of the square.
Notes / working: A square-based pyramid has 1 square face and 4 triangular faces.
The usual net: a square in the middle with one triangle on each of its four sides (like a star).
Other nets also work, for example four triangles joined in a row (alternately pointing up and down) with the square attached to the base of one triangle.

17 **[1]**

Which sentence describes capacity?

- ☐ A) The amount of space that a solid object takes up
- ☐ B) The amount that a container can hold when it is full
- ☐ C) The total area of all the faces of a 3D shape
- ☐ D) The distance all the way around a 2D shape

Answer: B) The amount that a container can hold when it is full
Notes / working: Capacity is how much a container can hold when it is full (for example 2 litres of juice) — answer B.
A describes volume. C describes surface area. D describes perimeter.

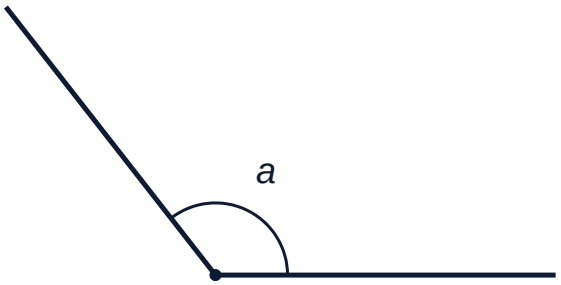
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18 Books are delivered to the library in a box shaped like a cuboid. It is 30 cm long, 20 cm wide and 10 cm high. **[2]**

Work out the surface area of the box. Show your working.

Answer: 2200 cm^2
Notes / working: A cuboid has three pairs of equal faces.
Top and bottom: 30 × 20 = 600, two of them = 1200
Front and back: 30 × 10 = 300, two of them = 600
Two ends: 20 × 10 = 200, two of them = 400
Surface area = 1200 + 600 + 400 = **2200 cm^2**.
Mark scheme: M1 for at least two of 600, 300, 200 (or 1200, 600, 400) seen; A1 for 2200. Common mistake: 1100 (each face counted once) or 6000 (that is the volume).

19 **[1]**



Estimate the size of angle a.

- ☐ A) 128°
- ☐ B) 38°
- ☐ C) 52°
- ☐ D) 152°

Answer: A) 128°
Notes / working: Angle a is **obtuse**: it is more than a right angle (90°) but less than a straight line (180°). So 38° and 52° (acute) are wrong.
It is about a right angle plus 40° (90° + 40° = 130°), and still far from a straight line, so **128°** (A) is the best estimate. 152° would be much flatter, close to a straight line.

20 **[1]**

Hassan wants to draw a circle with a diameter of 9 cm using compasses.
How far apart should he set the point and the pencil of the compasses? Explain your answer.

Answer: 1 mark: 4.5 cm with a reason, e.g. the compasses draw the radius / the radius is half of the diameter (9 ÷ 2 = 4.5). Do not accept 9 cm.
Notes / working: The distance between the compass point and the pencil is the **radius**.
Radius = diameter ÷ 2 = 9 ÷ 2 = **4.5 cm**.
In the real test you may be asked to draw the circle: put the point on the centre, keep the setting at 4.5 cm and turn the compasses all the way round.

21 At the zoo shop there is a game with a spinner showing four animals. Mona spins it 100 times. **[1]**
These are her results.

Animal	lion	zebra	monkey	giraffe
Frequency	12	25	9	54

(a) Mona's friend says, "Each animal is equally likely."
Do Mona's results support this? Explain your answer.

Answer: 1 mark: No, with a reason using the results, e.g. if each animal were equally likely each would come up about 25 times out of 100, but giraffe came up 54 times and monkey only 9 times. Do not accept "No" with no reason.
Notes / working: If the four animals were equally likely, each would land about 100 ÷ 4 = 25 times. Giraffe came up 54 times (more than half) and monkey only 9 times. The results are far from 25 each, so they **do not support** the statement — the spinner seems more likely to land on giraffe.

(b) Why is it better to spin the spinner 100 times than 10 times? **[1]**

- ☐ A) It makes the spinner fair
- ☐ B) Each animal will then come up the same number of times
- ☐ C) The results give a more reliable estimate of the probability
- ☐ D) The spinner will land on giraffe every time

Answer: C) The results give a more reliable estimate of the probability
Notes / working: With only a few trials the results can be very different from what we expect, just by chance. A **large number of trials** gives a more reliable estimate of the probability (C). More spins cannot change whether the spinner is fair (A), and the results will not become exactly equal (B).

22 At the zoo shop there is a game with a spinner showing four animals. Mona spins it 100 times. **[1]**
These are her results.

Animal	lion	zebra	monkey	giraffe
Frequency	12	25	9	54

Mona spins the spinner another 200 times. About how many times should she expect it to land on zebra?

Answer: 50 times
Notes / working: Zebra came up 25 times out of 100 (one quarter).
200 is double 100, so expect about 25 × 2 = **50 times**.

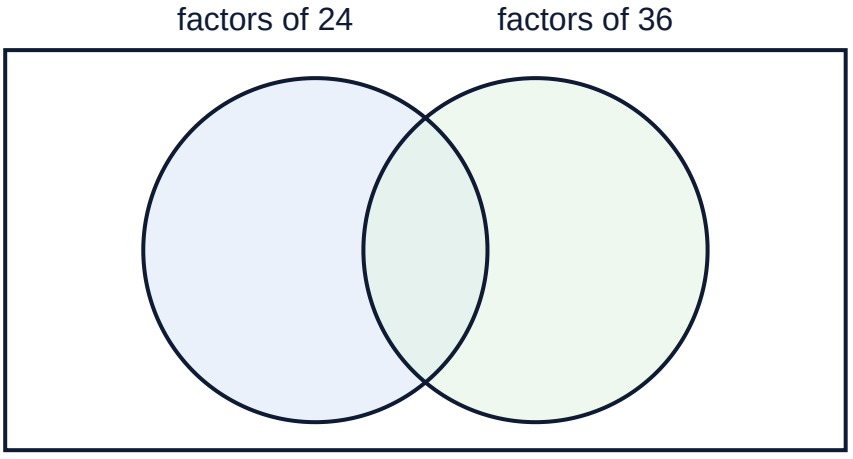
23 Pupils collect different kinds of data on sports day. **[1]**

Which of these is continuous data?

- ☐ A) the number of medals each class wins
- ☐ B) each pupil's favourite sport
- ☐ C) the number of pupils in each race
- ☐ D) the time each pupil takes to run 100 m

Answer: D) the time each pupil takes to run 100 m
Notes / working: Continuous data is measured and can take any value in a range (time, length, mass, temperature).
Time to run 100 m (for example 15.42 s) is continuous — **D**.
Numbers of medals and pupils are discrete (counted). Favourite sport is categorical.

24 **[1]**



Write the numbers 5, 8, 9, 12 and 18 in the correct places on the Venn diagram.

Answer: 1 mark for all five correct: 8 in 'factors of 24' only; 9 and 18 in 'factors of 36' only; 12 in the overlap; 5 outside both circles (inside the rectangle).
Notes / working: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36.
8 → factors of 24 only.
9 and **18** → factors of 36 only.
12 → a common factor, so it goes in the overlap.
5 → a factor of neither, so it goes outside both circles.

25 7 litres of juice are shared equally between 4 jugs. **[1]**

How many litres are in each jug? Give your answer as a mixed number or a decimal.

Answer: 1.75 or 7/4 or 1 3/4 litres
Notes / working: 7 ÷ 4 = 7/4 (a fraction is a division of the numerator by the denominator).
7/4 = **1 3/4 litres** = 1.75 litres.

26 A sequence starts at 2.5 and goes up by 0.75 each time. **[1]**
2.5, 3.25, 4, 4.75, …

Is 10 a term in this sequence? Explain how you know.

Answer: 1 mark: Yes, with a correct reason, e.g. (10 − 2.5) ÷ 0.75 = 10 steps, a whole number / it is the 11th term / continuing the sequence 5.5, 6.25, 7, 7.75, 8.5, 9.25, 10. Do not accept "Yes" with no working.
Notes / working: From 2.5 to 10 is 10 − 2.5 = 7.5.
7.5 ÷ 0.75 = 10, a whole number of steps, so **yes**, 10 is in the sequence.
It is 10 steps after the first term, so it is the **11th term**.
Check by counting on: 4.75, 5.5, 6.25, 7, 7.75, 8.5, 9.25, 10 ✓